PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Artificial Intelligence in Business and Marketing

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Submitted by:

Mandeep Kaur

2210991879

Supervised By:

Ms. Simar Virk

Mentor

Chitkara University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Artificial Intelligence in Business and Marketing"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of . The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura **Mandeep Kaur**

Date: 28.04.26 2210991879

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Ms. Simar Virk

Mentor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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ABSTRACT

This project explores the growing role of Artificial Intelligence (AI) in business and marketing. It highlights how AI enhances efficiency, personalization, and decision-making while also identifying its limitations in creativity and emotional engagement.

The study is based on secondary data sources and machine learning models applied to the Online Shoppers Purchasing Intention dataset. A comparative analysis between AI-driven, human-driven, and hybrid marketing approaches is conducted.

The findings suggest that while AI improves speed and accuracy, human creativity remains essential. The hybrid approach provides the best results in terms of engagement, conversion, and brand recall.

1. INTRODUCTION

Artificial Intelligence has become a transformative force in modern business and marketing. Companies are increasingly using AI technologies such as machine learning, Natural Language Processing (NLP), and recommendation systems to automate tasks and improve customer experiences. AI helps businesses analyze large datasets, predict customer behavior, and deliver personalized marketing campaigns. However, marketing is not only about efficiency - it also involves creativity, storytelling, and emotional connection. This creates a challenge: while AI improves speed and accuracy, it lacks originality and emotional depth. This project focuses on understanding this balance and evaluating the effectiveness of AI in marketing.

2. METHODOLOGY

2.1 Research Design

The research adopts a **comparative, analytical, and interpretative design** to evaluate the effectiveness of Artificial Intelligence in marketing. The study focuses on comparing three distinct marketing approaches:

- AI-driven marketing
- Human-led marketing
- Hybrid marketing (combination of AI and human effort)

The objective is to identify patterns, relationships, and performance differences among these approaches based on key marketing metrics such as engagement, conversion, and creativity. This design is suitable because it allows a structured comparison of different strategies while also interpreting their real-world implications in business environments.

2.2 Data Sources

The study is entirely based on **secondary data**, collected from credible and widely accepted sources to ensure reliability and validity. The major data sources include:

- Published research papers from platforms such as IEEE Xplore and Google Scholar
- Industry reports from organizations like McKinsey, Deloitte, Salesforce, HubSpot, and Statista
- Case studies of companies using AI in marketing
- The *Online Shoppers Purchasing Intention* dataset

These sources provide both **quantitative data (metrics, statistics)** and **qualitative insights (trends, observations)**, enabling a comprehensive analysis.

2.3 Data Selection Criteria

To maintain the quality and relevance of the research, specific criteria were followed while selecting data:

- Only recent studies (within the last 5–7 years) were included to reflect current technological advancements
- Preference was given to studies comparing AI-based and traditional marketing approaches
- Industry reports containing measurable performance indicators were prioritized
- Data sources were selected based on credibility, authenticity, and citation frequency

This filtering ensured that the research is based on **accurate, updated, and relevant information**.

2.4 Research Approach

The research follows a **multi-step analytical approach**:

- Identification of key performance indicators (KPIs) in marketing
- Collection of relevant data from secondary sources
- Creation of a synthesized dataset combining multiple reports
- Comparative analysis of AI-driven, human-led, and hybrid strategies
- Interpretation of results to derive meaningful insights

The approach focuses on understanding how different marketing strategies perform in terms of efficiency, engagement, creativity, and overall effectiveness.

2.5 Evaluation Parameters

The effectiveness of different marketing strategies was evaluated using the following parameters:

1. Operational Efficiency

Measures the time saved, cost reduction, and automation benefits achieved through AI.

2. Customer Engagement

Analyzes user interaction levels, click-through rates, and personalization effectiveness.

3. Conversion Performance

Evaluates the ability of marketing strategies to convert potential customers into actual buyers.

4. Creative Uniqueness

Assesses originality, innovation, and differentiation in marketing content.

5. Emotional Impact / Brand Recall

Measures how well the marketing content connects emotionally and remains memorable.

6. Accuracy

Determines the precision of predictions, targeting, and recommendations generated by AI.

7. Scalability

Evaluates the ability to expand marketing campaigns to larger audiences efficiently.

8. Flexibility and Adaptability

Measures how well strategies adjust to changing market trends and customer preferences.

9. Cost-effectiveness

Analyzes the return on investment and overall profitability of marketing strategies.

These parameters provide a **holistic framework** for comparing different approaches.

2.6 Machine Learning Models and Dataset

To support the research with practical analysis, machine learning models were applied to the **Online Shoppers Purchasing Intention dataset**, which contains customer behavior data.

Dataset Details

- Total records: 9,864
- Input features: 18
- Target variable: Purchase (Yes/No)

Models Used

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest

Model Performance

- Logistic Regression: 82% accuracy

- SVM: 84% accuracy
- Decision Tree: 86% accuracy
- Random Forest: 89% accuracy

The **Random Forest model** was selected as the best-performing model due to its higher accuracy and ability to reduce overfitting.

2.7 Limitations of Methodology

Despite careful design, the study has certain limitations:

- Reliance on secondary data may introduce inconsistencies
- Lack of primary data limits control over variables
- Differences in methodologies across sources may affect uniformity
- Synthesized dataset may contain minor approximations

These limitations are acknowledged to maintain transparency in the research.

3. TOOLS AND TECHNOLOGIES

The project utilizes a combination of data analysis tools and AI technologies to conduct the research effectively.

Programming Language

- Python (for machine learning and data analysis)

Machine Learning Techniques

- Classification algorithms (Logistic Regression, SVM, Decision Tree, Random Forest)

AI Technologies Used

- Natural Language Processing (NLP) for content analysis
- Recommendation systems for personalization

Data Handling Tools

- Data preprocessing techniques
- Statistical analysis methods

These tools enable efficient data processing, model training, and result evaluation.

4. IMPLEMENTATION

The implementation of the project was carried out in multiple structured phases:

Phase 1: Data Collection

Relevant data was gathered from research papers, industry reports, and datasets.

Phase 2: Data Preparation

The collected data was cleaned, filtered, and organized to remove inconsistencies and ensure usability.

Phase 3: Dataset Synthesis

A combined dataset was created by integrating insights from multiple sources to allow comparative analysis.

Phase 4: Model Application

Machine learning models were applied to the dataset to analyze customer behavior and predict purchase intentions.

Phase 5: Performance Evaluation

Each model was evaluated based on accuracy, and the best-performing model (Random Forest) was selected.

Phase 6: Comparative Analysis

AI-driven, human-led, and hybrid marketing strategies were compared using defined evaluation parameters.

Phase 7: Result Interpretation

The findings were analyzed to understand the strengths and limitations of AI in marketing.

5. MAJOR FINDINGS / RESULTS

5.1 Efficiency Analysis

AI-driven marketing campaigns significantly reduce the time required for campaign setup and execution. However, faster execution does not always guarantee better performance.

5.2 Engagement and Conversion Trends

Human-led campaigns demonstrate stronger engagement due to emotional storytelling, while AI campaigns rely more on data-driven targeting. Hybrid campaigns achieve the best results by combining both.

5.3 Creativity Analysis

AI-generated content often lacks originality and tends to follow repetitive patterns. Human marketers excel in creativity, while hybrid approaches achieve the highest creativity scores.

5.4 Accuracy vs Effectiveness

Although AI systems achieve high prediction accuracy, they do not always translate into better emotional connection or brand recall.

5.5 Key Insight

The study clearly indicates that:

- AI is best for efficiency and accuracy
- Humans are best for creativity and emotional connection
- Hybrid approach delivers the best overall performance

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The study highlights that Artificial Intelligence is transforming marketing by improving efficiency, scalability, and data-driven decision-making. However, it also reveals that AI lacks the human qualities of creativity, emotional intelligence, and storytelling.

Therefore, AI should not be seen as a replacement for human marketers but as a **supporting tool** that enhances their capabilities. The most effective strategy is a hybrid approach that combines AI's analytical power with human creativity.

6.2 Future Scope

The research can be extended in several ways:

- Incorporating real-time datasets for more accurate analysis
- Exploring advanced generative AI tools for creative marketing
- Conducting primary research through surveys and experiments
- Studying industry-specific applications of AI in marketing
- Improving AI models to enhance creativity and emotional intelligence

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APPENDICES

Appendix A: Dataset Description

The dataset used in this research is the **Online Shoppers Purchasing Intention Dataset**, which contains information about customer behavior during online shopping sessions.

Key Features of Dataset:

- Total records: 9,864
- Number of input features: 18
- Target variable: Purchase (Yes/No)
- Features include:
    - Bounce Rate
    - Exit Rate
    - Page Value
    - Product Page Visits
    - Visitor Type
    - Administrative and Informational Pages

This dataset helps in analyzing customer engagement and predicting purchase behavior.